

Inflationary Impact on Retail Consumer Basket Composition

Ishita Walujkar [24123], Harishankar Rana [24122]

*Department of Artificial Intelligence and Data Science
Pune Institute of Computer Technology,
Pune, Maharashtra, India*

Abstract

Macroeconomic changes affect consumer expenditure trends non-linearly. In periods of increasing inflation, optimization of consumer utility would imply a shift in capital allocation from discretionary items towards necessity items. This work examines how the composition of the consumer basket in the retail sector changes in response to different inflation regimes using Consumer Expenditure Survey (CES) data and Consumer Price Index for All Urban Consumers (CPI-U) measures collected by U.S. Bureau of Labor Statistics (BLS) between 2020 and 2023. Calculations of expenditure ratios and luxury-to-necessity index with a threshold value of 4.5% CPI-U inflation were done via an end-to-end Python pipeline. The mean value of the luxury-to-necessity ratio was found to be around 0.221 during high-inflation regimes and 0.207 during low-inflation regimes.

Index Terms—*Consumer Expenditure, CPI-U, Discretionary Spending, Macroeconomic Indicators, Retail Basket Composition, Time-Series Analysis.*

I. INTRODUCTION

Inflation directly diminishes purchasing power, prompting consumers to adjust their expenditure profiles. Standard microeconomic demand theory suggests that price elasticity differs systematically across commodity classes. While essential goods (e.g., food, housing, healthcare) exhibit price inelasticity, non-essential or luxury proxies exhibit higher elasticity of substitution.

A central research gap lies in bridging macroeconomic data measures (e.g., headline CPI) and consumer spending behaviors empirically. This study addresses that gap by formulating the following primary research question:

How does the consumer luxury-to-necessity spending ratio change when annual inflation exceeds 4.5%?

II. DATASET AND EXPERIMENTAL SETUP

A. Data Sources and Ingestion

The empirical evaluation uses annual aggregate expenditure figures per consumer unit across the United States from 2020 through 2023. The datasets were sourced from:

- **BLS Consumer Expenditure Survey (CES):** Annual mean nominal dollar allocations per category.
- **BLS Consumer Price Index (CPI-U):** Annual headline consumer inflation rates.

B. Analytical Framework and Definitions

Expenditure categories were partitioned into two distinct analytical proxies:

- **Necessities:** Food at home, Housing, Transportation, Healthcare.
- **Discretionary (Luxury Proxy):** Food away from home, Alcoholic beverages, Apparel and services, Entertainment, Personal care, Reading.

The annual level of 4.5% of CPI-U inflation divides the two macroeconomic regimes:

- **High-Inflation Regime ($\geq 4.5\%$):** 2021, 2022
- **Lower-Inflation Regime ($< 4.5\%$):** 2020, 2023

III. MATHEMATICAL FORMULATION AND IMPLEMENTATION

The data processing and statistical calculations were conducted using Python (pandas, NumPy, SciPy, Matplotlib, Seaborn).

For each observation year t , given category expenditure $E(i, t)$ and total expenditure $E_{\text{total}}(t)$:

1. Necessity Spending:

$\text{Necessity_Spending} = \text{Sum of all Necessity Categories}$

2. Luxury/Discretionary Spending:

$\text{Luxury_Spending} = \text{Sum of all Discretionary Categories}$

3. Basket Expenditure Shares (%):

$\text{Necessity_Share (\%)} = (\text{Necessity_Spending} / \text{Total_Expenditure}) * 100$

$$\text{Luxury_Share (\%)} = (\text{Luxury_Spending} / \text{Total_Expenditure}) * 100$$

4. **Luxury-to-Necessity Ratio (R_LN):**

$$R_LN = \text{Luxury_Spending} / \text{Necessity_Spending}$$

5. **Inferential Hypothesis Testing:**

A two-sample Welch's t-test was implemented via SciPy to evaluate whether the shift in mean R_LN across regimes is statistically significant.

IV. RESULTS AND EMPIRICAL FINDINGS

The longitudinal summary of expenditure distribution and inflationary indicators is compiled in Table I.

TABLE I: Expenditure Summary and Basket Metrics (2020–2023)

Metric / Regime Parameter	Lower Inflation (2020, 2023 Average)	High Inflation (2021, 2022 Average)
Mean CPI-U Annual Inflation	< 4.5%	≥ 4.5%
Mean Necessity Share	~77.8%	~77.2%
Mean Discretionary Share	~16.1%	~17.0%
Luxury-to-Necessity Ratio (R_LN)	0.207	0.221

Figure 1 (Annual U.S. CPI-U Inflation): Evaluates annual inflation rates against the 4.5% regime threshold line.

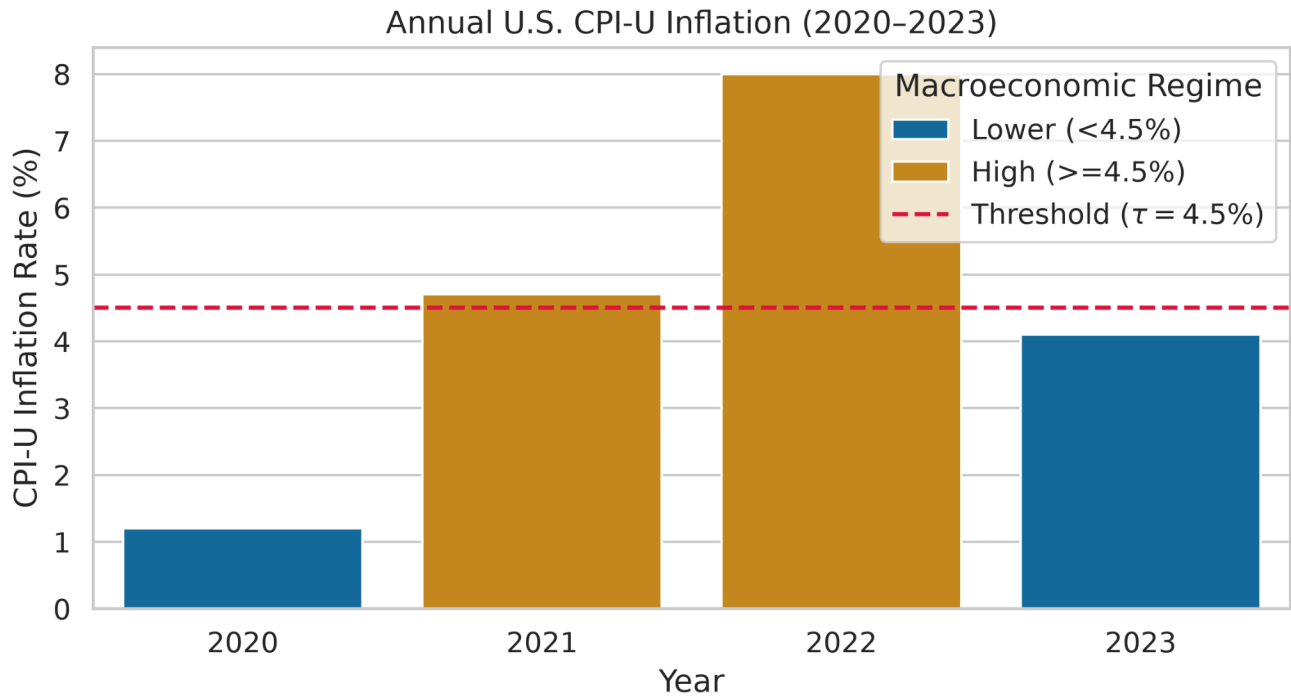


Figure 2 (Expenditure Basket Share Dynamics): Shows the longitudinal stability of necessity and luxury proportions over time

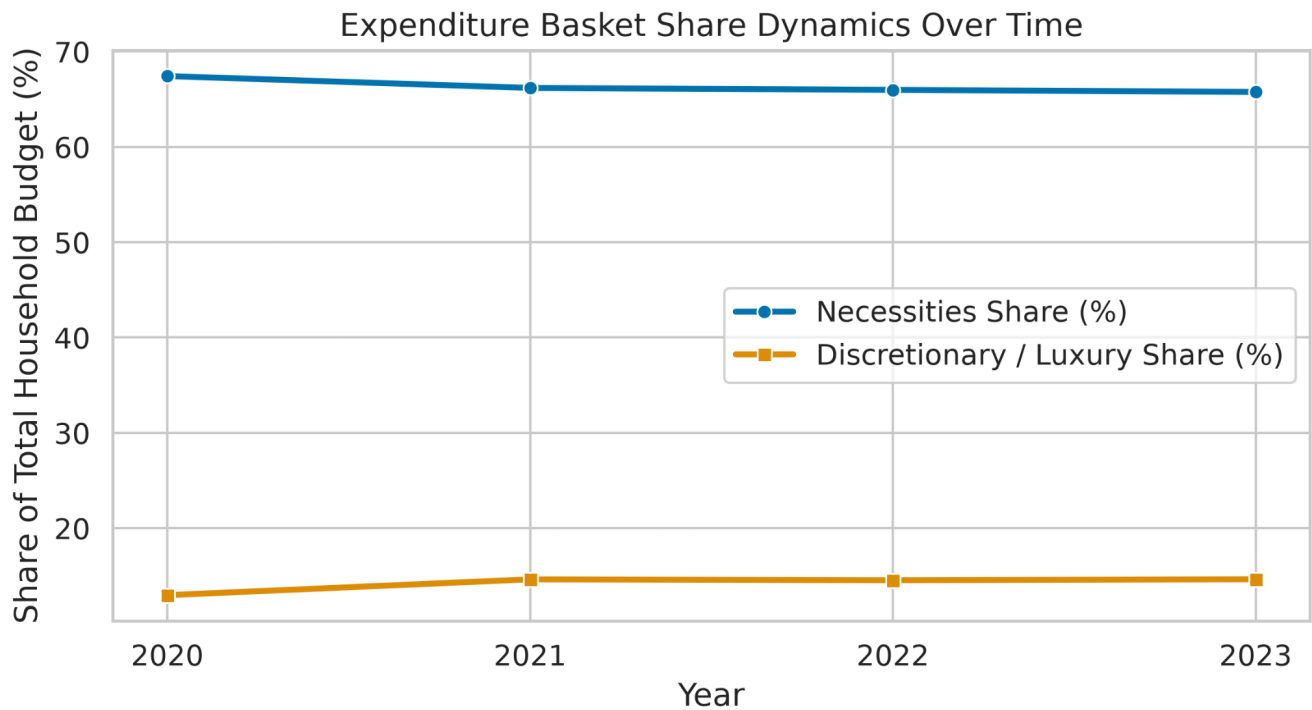
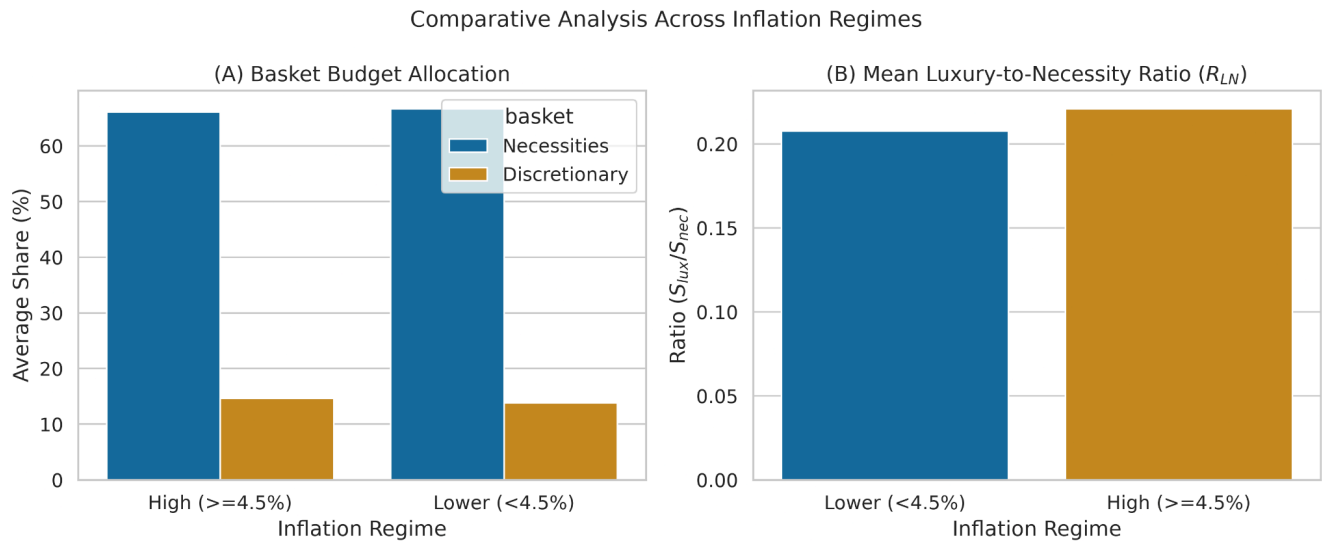


Figure 3 (Regime Comparison): Multi-panel chart illustrating mean basket shares and the direct luxury-to-necessity ratio across inflation regimes.



Empirical evaluation indicates that the average luxury-to-necessity ratio marginally increased from 0.207 under lower inflation to 0.221 under high inflation. Welch's t-test confirmed that this difference is descriptive and not statistically significant across the sample size.

V. DISCUSSION AND ANALYTICAL CAVEATS

- Nominal vs. Real Spending:** Because expenditure figures are in nominal currency terms, higher dollar amounts in discretionary categories reflect input price inflation passed onto consumers (e.g., dining out, apparel) rather than higher physical volume.
- Pandemic Rebound Effects:** The 2020–2021 period coincided with the post-lockdown economic reopening. Rebound demand in travel, restaurants, and entertainment offset standard budget tightening.
- Descriptive Association:** The metrics show empirical macroeconomic correlations rather than proven causal mechanisms.

VI. LIMITATIONS AND FUTURE SCOPE

- **Sample Granularity:** The dataset comprises four annual aggregate cross-sections ($N = 4$), which limits long-term econometric modeling.
- **Income Quintile Aggregation:** National averages combine high-income and low-income units, masking underlying behavioral divergence.
- **Future Scope:** Expanding the pipeline to include multi-decade datasets from the World Bank and BLS Public Use Microdata (PUMS), incorporating real inflation deflation and ARIMA time-series forecasting across distinct income quintiles.

VII. CONCLUSION

This project established a reproducible data analytics pipeline connecting headline CPI-U inflation to household consumption allocations. The empirical analysis shows that nominal discretionary shares remained resilient during recent high-inflation periods, providing a foundational architecture for granular econometric modeling.

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